

Roll No.

7574

**B. C. A. (Second Semester)
EXAMINATION, 2025-26
Paper I**

**COMPUTER SYSTEM ARCHITECTURE
Course Code : B230201T**

Time : Two Hours }

[Maximum Marks : 75

Note : This question paper is divided into three Sections—A, B and C.
Attempt questions from all Sections as directed.

Inst. : The candidates are required to answer only in serial order. If there are many parts of a question, answer them in continuation.

Section—A

Very Short Answer Type Questions

Note : Attempt any *three* questions. Give answer of each question in about 50 words. Each question carries 3 marks. $3 \times 3 = 9$

1. (a) Name different types of computer register and buses.
(b) Define stack and stack pointer in brief.
(c) What is Register Transfer Language ?
(d) Discuss in brief Instruction Format.
(e) Explain the function of Cache memory in brief.

Section—B

Short Answer Type Questions

Note : Attempt any *four* questions. Answer each question in about 225 words. Each question carries 9 marks. $4 \times 9 = 36$

2. (a) Explain the working of DMA with the help of neat diagram.
(b) Explain the concept of instruction cycle and its different phases.

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- (c) What do you mean by Addressing modes ? Explain in brief its types with example.
- (d) Explain stored program design with neat diagram.
- (e) Differentiate between RISC and CISC.
- (f) Explain common bus system in a basic computer with diagram.
- (g) Explain three instruction code format related to memory, register and input-output.

Section—C

Long Answer Type Questions

Note : Attempt any *two* questions. Answer each *question* in about 475 words. Each question carries 15 marks. $2 \times 15 = 30$

3. (a) Define computer organization in detail with the help of diagram.
- (b) Explain virtual memory with the help of diagram.
- (c) Explain micro-operation and also explain logic micro-operation in detail, with the help of neat diagram.
- (d) Explain the working of general registers organizations and each block also, with the help of diagram.